

FUTURE STACK

A Full Stack Project Report

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Introduction :

Step into the future of career planning and opportunity management with FutureStack, a modern web application crafted using the power of React.js. Designed for students and early career professionals, FutureStack brings clarity and structure to the overwhelming process of tracking internships, hackathons, deadlines, and application progress.

Blending clean design with user centric workflows, the platform turns scattered opportunities into a unified, interactive dashboard that keeps you organized and future-ready. Whether you are managing five opportunities or fifty, FutureStack ensures that every detail, deadline, and progress update remains accessible and easy to maintain.

Created with a focus on simplicity, responsiveness, and real-world usability, FutureStack harnesses the capabilities of React, Tailwind CSS, Axios, and JSON Server to deliver a seamless tracking experience. From visual status boards to calendar based deadline highlights, the system transforms opportunity management into an intuitive and engaging process.

Say goodbye to messy spreadsheets and missed deadlines. FutureStack is your structured companion for staying on top of every application and making informed steps toward your future.

Scenario-Based Intro :

Imagine a typical day for a student preparing for placements or upcoming hackathons. You sit down with your laptop, determined to sort out the opportunities you want to apply for. Instead of sifting through bookmarks, notes, screenshots, or random reminders, you open your organized digital assistant, FutureStack.

As soon as the dashboard loads, FutureStack shows you everything you need. At a glance, you see how many opportunities you have applied to, which ones are shortlisted, which deadlines are approaching soon, and which tasks need your attention today.

You click on an opportunity card and instantly view its details: the link, notes, deadline, and current status. Changing the status is effortless. Need to update it to “Interviewed”? Just a quick selection and it moves to the right column on the status board.

Later, while planning your week, you switch to the calendar view. Important dates appear clearly marked with color-coded indicators for internships and hackathons. You instantly see what's due tomorrow or next week, helping you stay prepared and confident.

By the end of your session, you export your full opportunity history as a PDF to share with your mentor. In a few seconds, you walk away with a clean, structured, downloadable summary of your progress. That is the experience FutureStack is built to provide - clarity, structure, and control.

Target Audience :

FutureStack is designed specifically for:

1. Students Preparing for Internships & Placements

Those who actively search for internships, jobs, hackathons, and academic opportunities, and need a structured system to manage them.

2. Early Career Professionals

Individuals in the early stages of their careers who want to track applications, deadlines, interview rounds, and progress in an organized way.

3. Hackathon & Project Enthusiasts

Students participating in multiple competitions who need a clear overview of deadlines, categories, and submissions.

4. Mentors & Career Guides

Mentors who review students' applications can use exported PDFs to quickly assess progress and provide guidance.

5. Anyone Managing Multiple Opportunities

Anyone who struggles with scattered notes, reminders, and browser bookmarks can use FutureStack as a unified dashboard for better clarity and control.

Project Goals and Objectives :

The primary goal of FutureStack is to provide a centralized, intuitive, and efficient platform that simplifies the process of tracking internships, hackathons, and application progress. To achieve this, the project focuses on delivering a structured system that helps users stay organized, meet deadlines, and make informed decisions throughout their career journey.

FutureStack aims to:

1. Centralize Opportunity Management

Combine scattered information from notes, spreadsheets, and bookmarks into a single, well-organized dashboard.

2. Improve Productivity and Time Management

Provide visual reminders, upcoming deadlines, and progress indicators to help users prioritize their tasks effectively.

3. Enhance the User Experience with Simple, Modern UI

Deliver a clean, responsive, and user-friendly interface that encourages consistent usage and reduces complexity.

4. Offer Clear Progress Insights

Provide analytics such as applied, shortlisted, selected, and rejected counts to help users evaluate their career preparation.

5. Support Informed Planning Through Visualization

Use a calendar view and Kanban-style status board to give users immediate clarity about their upcoming deadlines and application stages.

6. Enable Easy Search, Filtering, and Documentation

Allow users to quickly find opportunities and export their progress as a PDF for mentor reviews or record-keeping.

7. Maintain Scalability and Modularity

Use React components, API service layers, and reusable design patterns to ensure the application is easy to maintain and expand in the future.

Key Features :

1. Opportunity Management (CRUD System)

FutureStack offers a complete Create, Read, Update, and Delete workflow for managing your internships and hackathons. Each opportunity stores key information such as title, description, link, deadline, category, status, and notes. This ensures that all details remain well organized and accessible whenever needed.

2. Status Board (Kanban View)

A visually structured kanban-style board helps you understand your progress instantly. Opportunities automatically appear under categories such as Applied, Shortlisted, Interviewed, Selected, or Rejected. Changing status is simple and updates all views across the application.

3. Dashboard with Insights

The dashboard provides meaningful statistics - total opportunities, applied count, shortlist count, selected count and highlights upcoming deadlines. The clean layout helps you make informed decisions about what to prioritize next.

4. Calendar-Based Deadline Visualization

FutureStack integrates a calendar view that highlights important deadlines using color-coded indicators. Internships and hackathons are visually distinguishable, enabling smarter planning at a glance.

5. Search and Filters

With built-in search and filtering, you can quickly find opportunities by title, description, category, or status. This feature ensures fast access to exactly what you need without scrolling endlessly.

6. PDF Export

FutureStack allows users to export a complete or filtered set of opportunities as a clean, formatted PDF. This is useful for progress reviews, mentor discussions, or personal documentation.

7. Responsive and Accessible Design

Built with Tailwind CSS, the interface adapts effortlessly across devices. Whether on a laptop, tablet, or mobile phone, FutureStack provides a consistent and user friendly layout.

8. Notifications and Error Handling

React Toastify is used to provide clear, instant feedback for actions like creation, updating, or deletion. Success and error messages guide the user smoothly through the experience.

PRE-REQUISITES :

Before developing and running FutureStack, the following prerequisites must be fulfilled:

🔗 **Node.js and npm:**

Node.js is a powerful JavaScript runtime environment that allows you to run JavaScript code on the local environment. It provides a scalable and efficient platform for building network applications.

Install Node.js and npm on your development machine, as they are required to run JavaScript on the server-side.

- Download: <https://nodejs.org/en/download/>
- Installation instructions: <https://nodejs.org/en/download/package-manager/>

● **React.js:**

React.js is a popular JavaScript library for building user interfaces. It enables developers to create interactive and reusable UI components, making it easier to build dynamic and responsive web applications.

Install React.js, a JavaScript library for building user interfaces.

- Create a new React app:

```
npm create vite@latest
```

Enter and then type project-name and select preferred frameworks and then enter

- Navigate to the project directory:

```
cd project-name
```

```
npm install
```

- Running the React App:

With the React app created, you can now start the development server and see your React application in action.

- Start the development server:

```
npm run dev
```

This command launches the development server, and you can access your React app at <http://localhost:5173> in your web browser.

- **HTML, CSS, and JavaScript:** Basic knowledge of HTML for creating the structure of your app, CSS for styling, and JavaScript for client-side interactivity is essential.
- **Version Control:** Use Git for version control, enabling collaboration and tracking changes throughout the development process. Platforms like GitHub or Bitbucket can host your repository.
 - Git: Download and installation instructions can be found at: <https://git-scm.com/downloads>
- **Development Environment:** Choose a code editor or Integrated Development Environment (IDE) that suits your preferences, such as Visual Studio Code, Sublime Text, or WebStorm.
 - Visual Studio Code: Download from <https://code.visualstudio.com/download>
 - Sublime Text: Download from <https://www.sublimetext.com/download>
 - WebStorm: Download from <https://www.jetbrains.com/webstorm/download>

Technology Stack :

Frontend: React.js

Styling: Tailwind CSS

Routing: React Router DOM

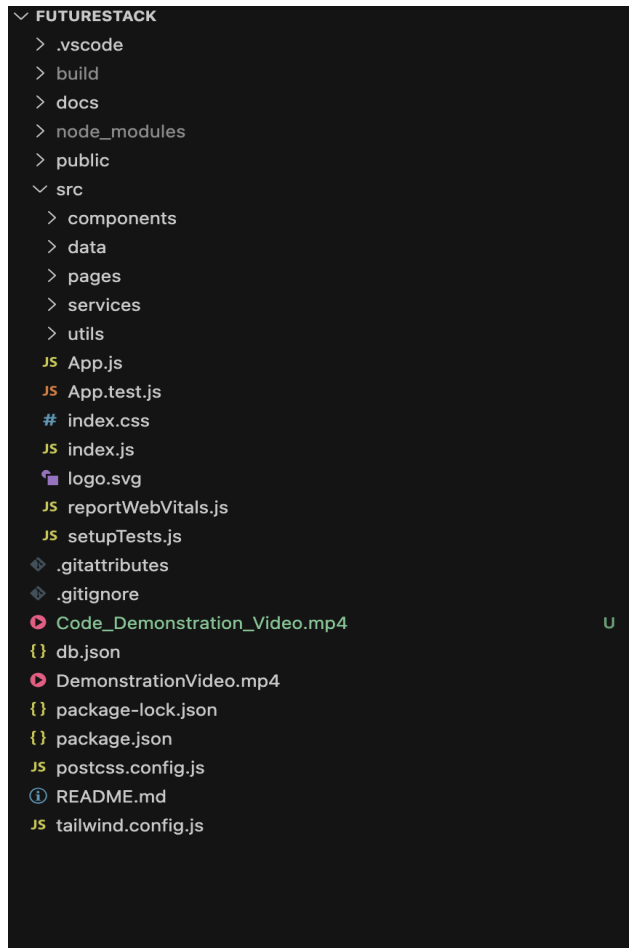
HTTP Requests: Axios

Backend (Mock Server): JSON Server

Additional Tools: React Calendar, React Icons, jsPDF, React Toastify

React components, modular design practices, reusable UI elements, and the separation of API services ensure scalability, maintainability, and clarity of the project structure.

Project structure :



- **App.js, index.js, index.css, App.test.js, reportWebVitals.js, setupTests.js:** Entry-point files wiring up React, global styles, testing utilities, and performance logging for the root application.
- **src/components/common/*.jsx:** Shared UI primitives (buttons, cards, modals, navbar) reused across pages to keep styling/behavior consistent.
- **DeadlineWidget.jsx, StatsCard.jsx:** Dashboard-only widgets presenting upcoming deadlines and KPI cards.
- **src/components/opportunities/*.jsx:** Opportunity-specific components (cards, forms, lists) encapsulating CRUD UI logic.
- **StatusCard.jsx, StatusColumn.jsx:** Kanban board building blocks used to visualize opportunities by status.
- **sampleData.js:** Example data used for mock rendering, testing, or documentation snippets.

- **src/pages/*.jsx plus Calendar.css:** Route-level pages (Dashboard, Lists, Status Board, Calendar, Reports, etc.) composing feature sections and hooking into services.
- **api.js:** Axios-based service layer centralizing HTTP configuration, interceptors, and REST helpers.
- **dateHelpers.js, pdfExport.js:** Pure utility modules for date calculations and jsPDF report generation.

PROJECT FLOW :

Project demo :

Let's take a look at Live deployment and Codebase of FutureStack

Live Website : <https://futurestack-tracker.vercel.app/>

Github : <https://github.com/Venkat-Kolasani/FutureStack>

Website Demonstration Video :  **DemonstrationVideo.mp4**

Code Demonstration Video :  **Code_Demonstration_Video.mp4**

Project Flow :

Milestone 1: Project Setup and Configuration

- Install required tools and software:
- Open the FutureStack folder, run npm install, and ensure Node.js ≥ 18 , Git, and VS Code are available.
- Libraries used: React 19, React Router DOM, Tailwind CSS, Axios, React Toastify, React Calendar, jsPDF, React Icons, JSON Server.

Reference docs:

- React install – <https://react.dev/learn/installation>
- Tailwind setup – <https://tailwindcss.com/docs/guides/create-react-app>
- React Router – <https://reactrouter.com/en/main/start/tutorial>

Milestone 2: Web Development

1. Setup React application:

- CRA bootstrap already in place; index.js hydrates `<App />` within StrictMode.
- App.js wraps AppContent, `<ToastContainer/>`, and router logic.
- Conditional navbar hides on `/`, uses `<Routes>` for Dashboard, Lists, StatusBoard, Calendar, Reports, CRUD forms.

2. Design UI components:

- components/common/* holds Button, Card, Modal, Navbar.
- Dashboard-specific widgets live in components/dashboard, Kanban cards in components/statusboard.

- Layout/styling powered by Tailwind utility classes defined inline; index.css adds global gradient background and font settings.
- Navigation handled by <Navbar/> plus React Router links.

3. Implement frontend logic:

- Each page fetches data via opportunityService (services/api.js) which wraps Axios.
- CRUD forms (pages/AddOpportunity.jsx, EditOpportunity.jsx) use local state, validation, and toasts for feedback.
- Status Board groups opportunities by status; Calendar maps deadlines to tiles; Reports invoke utils/pdfExport.js to generate PDFs.
- utils/dateHelpers.js centralizes date math for overdue/upcoming calculations.

Milestone 3: Data & Backend Layer

1. JSON Server setup:

- npm run server starts mock REST API at http://localhost:3001 reading db.json.
- Endpoints: /opportunities with GET/POST/PATCH/DELETE.

2. Data flow:

- Component action → opportunityService → Axios → JSON Server → db.json.
- Responses update component state; react-toastify displays success/error.

Milestone 4: Deployment & Demo

1. Build + Deploy:

- npm run build, deploy to Vercel

Website Flow :

- **User Entry → Landing Page**

Visitors load / to see the Home page. CTA routes them into the authenticated experience (Dashboard).

- **Navigation & Layout**

App.js renders Navbar (hidden on Home) and Routes. ToastContainer stays mounted for system-wide notifications.

- **Dashboard Initialization**

Dashboard.jsx fetches opportunities via opportunityService.getAll(). Statistics, upcoming deadlines, and alerts populate from the response.

- **Feature-Specific Routes**

- CRUD flows: AddOpportunity / EditOpportunity display OpportunityForm. Submission triggers create/update service calls, then redirects with success toasts.
- InternshipList / HackathonList filter opportunities by category with search + status filters.
- StatusBoard groups items by status, enabling quick status edits (PATCH requests).
- Calendar maps deadlines onto react-calendar; selecting a date reveals opportunities due that day.
- Reports uses generatePDF from utils/pdfExport to export summaries or full details.

- **Service Layer**

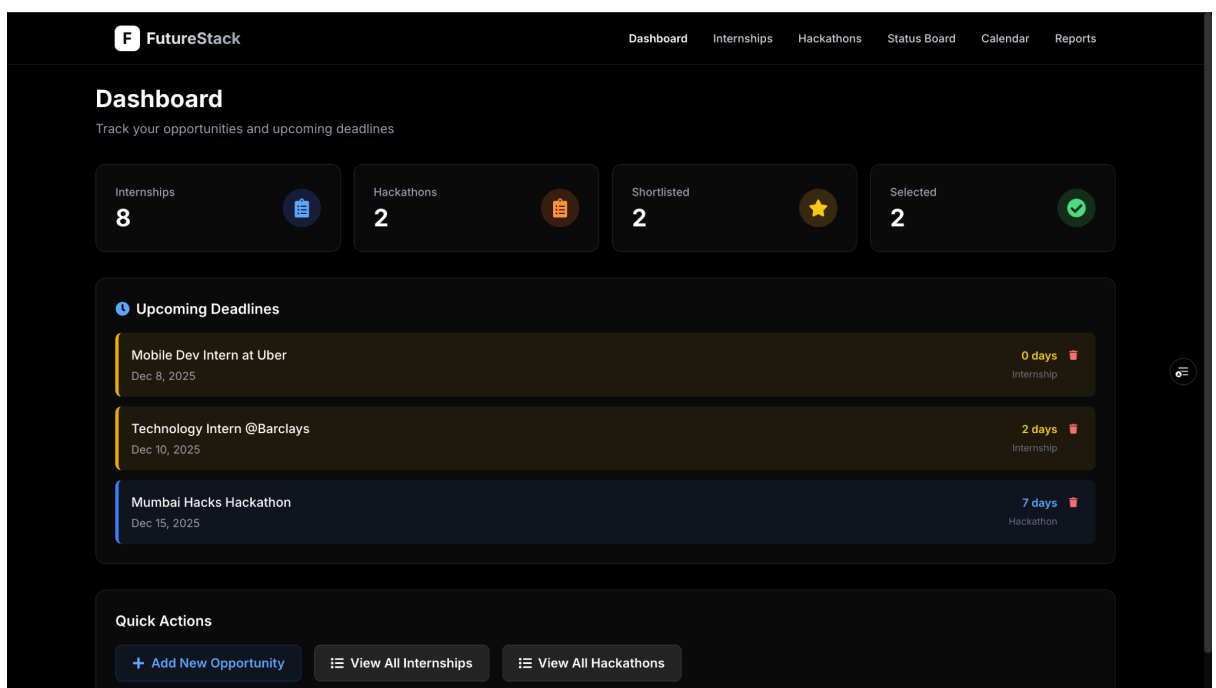
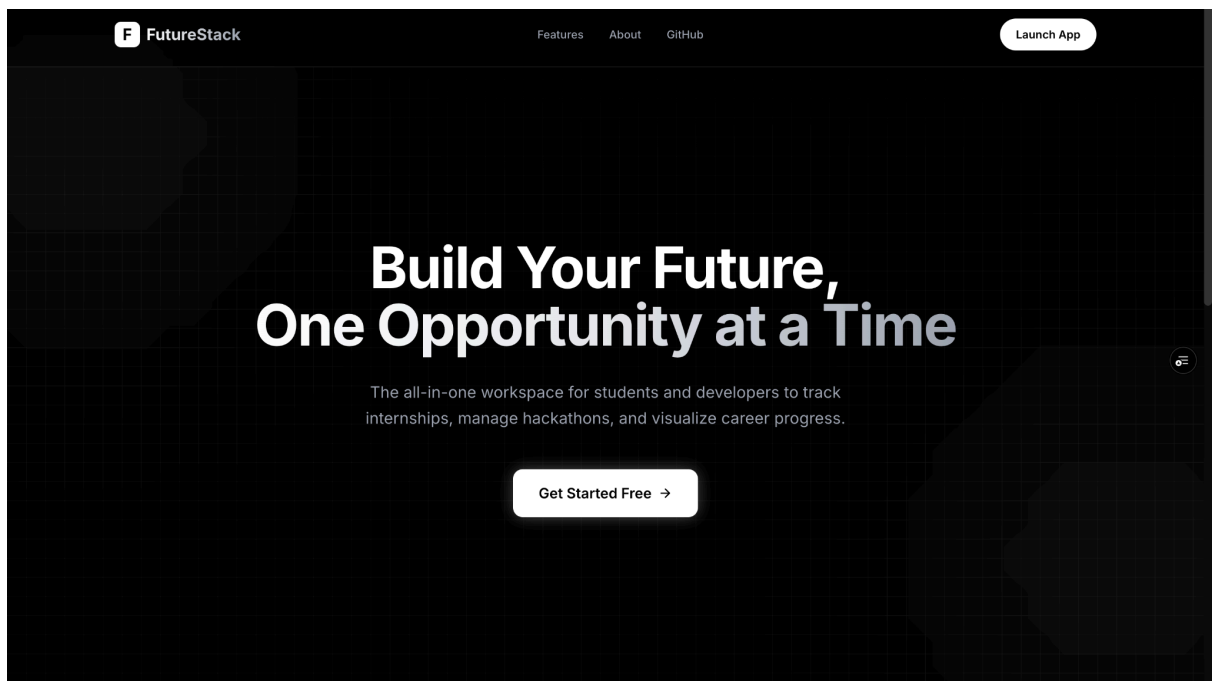
services/api.js centralizes Axios instance, interceptors, and REST helpers. All pages/components call this layer to read/write opportunities.

- **State Updates & Feedback**

Successful API responses update local state, re-render UI, and display react-toastify messages. Errors are caught, logged, and surfaced via toast.

Project Execution:

1. Install dependencies with `npm install` (or `yarn install`) after cloning the repo.
2. Start the mock backend by running `npm run server`; the JSON server listens on `http://localhost:3001`.
3. Launch the React frontend using `npm start` (or `npm run dev` if you adapt it to Vite). The app opens at `http://localhost:3000`.



FutureStack

DashboardInternshipsHackathonsStatus BoardCalendarReports

Internships

Manage and track your internship applications

Search by title or description...

All Statuses

Showing 8 of 8 internships

Full Stack Intern at Airbnb

internship

Work on host and guest experience features

Deadline: Dec 22, 2025
14 days remaining

Selected

View Opportunity

EditDelete

Mobile Dev Intern at Uber

internship

Develop features for Uber and Uber Eats apps

Deadline: Dec 8, 2025
0 days remaining

Applied

View Opportunity

EditDelete

Software Engineer Intern

internship

Build/Integrate tools to automate the process and improve the efficiency Create, maintain and improve...

Deadline: Feb 11, 2026
65 days remaining

Rejected

View Opportunity

EditDelete

Software Engineering, New Grad

internship

Preferred qualifications : >One or more areas of specialized knowledge balanced with general skills a...

Deadline: Dec 24, 2025
16 days remaining

Applied

View Opportunity

EditDelete

Microsoft Internship

internship

Deadline: Feb 6, 2026
60 days remaining

Applied

View Opportunity

EditDelete

Technology Intern @Barclays

internship

Deadline: Dec 10, 2025
2 days remaining

Shortlisted

View Opportunity

EditDelete

FutureStack

DashboardInternshipsHackathonsStatus BoardCalendarReports

Status Board

Track your application progress across different stages

Applied3

Mobile Dev Intern at Uber

internship

Deadline: Dec 8, 2025

Update Status

Applied

Microsoft Internship

internship

Deadline: Feb 6, 2026

Update Status

Applied

React Intern in SmartBridge

internship

Deadline: Jan 10, 2026

Update Status

Applied

Shortlisted2

Software Engineering, New Grad

internship

Deadline: Dec 24, 2025

Update Status

Shortlisted

Technology Intern @Barclays

internship

Deadline: Dec 10, 2025

Update Status

Shortlisted

Interviewed1

Machine Learning Intern @ SmartBridge

internship

Deadline: Jan 1, 2026

Update Status

Interviewed

Selected2

Full Stack Intern at Airbnb

internship

Deadline: Dec 22, 2025

Update Status

Selected

Mumbai Hacks Hackathon

hackathon

Deadline: Dec 15, 2025

Update Status

Selected

Rejected

CalHacks

hackathon

Deadline: J

Update Statu

Rejected

Software

internship

Deadline: F

Update Statu

Rejected

18

F

FutureStack

Dashboard

Internships

Hackathons

Status Board

Calendar

Reports

Calendar

View all your deadlines in calendar format

●

 Internship Deadline

●

 Hackathon Deadline

📅

 Click on a date to view opportunities

«

<

December 2025

>

»

MON	TUE	WED	THU	FRI	SAT	SUN
1	2	3	4	5	6	
7	8	9	10	11	12	
13	14	15	16	17	18	
19	20	21	22	23	24	
25	26	27	28	29	30	
31	1	2	3	4		

Total Deadlines

Internship Deadlines

Hackathon Deadlines

F

FutureStack

Dashboard

Internships

Hackathons

Status Board

Calendar

Reports

Reports & Export

Generate and download PDF reports of your opportunities

📄

 Export Options

●

 All Opportunities

●

 Selected Opportunities

●

 Summary Only

📄

 Download PDF

Statistics

Total:

10

Applied:

3

Shortlisted:

2

Interviewed:

1

Selected:

2

Rejected:

2

Preview

CalHacks

Category: Hackathon

Status: Rejected

Deadline: Jan 8, 2026

UC Berkeley's largest collegiate hackathon

Notes: Working on healthcare project idea

Full Stack Intern at Airbnb

Category: Internship

Status: Selected

Deadline: Dec 22, 2025

Work on host and guest experience features

Notes: Offer received! Negotiating start date

Mobile Dev Intern at Uber

Category: Internship

Status: Applied

Deadline: Dec 8, 2025

Develop features for Uber and Uber Eats apps

Notes: Received rejection email, will apply again next year

Software Engineer Intern

Category: Internship

Status: Applied

Deadline: Dec 15, 2025

Develop new features for our web application

Notes: Interview scheduled for Dec 10, 2025

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Conclusion:

FutureStack is designed to bring structure and efficiency to student career planning. With its blend of intuitive UI, modern React based architecture, and practical features like status tracking, calendar visualization, search and filtering, and PDF export, it acts as a complete personal dashboard for managing internships and hackathons.

By simplifying the entire opportunity tracking process, FutureStack empowers students to stay organized, meet deadlines, and move forward with confidence in their career journey

Website Demo Link : <https://futurestack-tracker.vercel.app/>